Personal, Relevant Background, and Future Goals Statement

Motivation and Background

I was about eight when I asked my grandpa what his favorite molecule was. He was a PhD chemist, so I expected some complicated compound I'd struggle to pronounce. Instead, he simply said, "Oh, that's easy, water!" At the time, this answer confused me. Water was everywhere. It was just a standard part of life, mundane even. "You'll be surprised how even the most simple things become the most beautiful once you truly understand them," he explained. Curious to test his statement, I took apart my dad's car radio, hoping to figure it out and see its beauty. But, instead, I was left with a broken radio and an upset father. Nonetheless, I continued my search, delving into books on famous inventors and scientists. At the time, I was most drawn to Isaac Newton. Not just because we shared a name, but mainly because I was fascinated with his discoveries in optics. Learning that white light was composed of a continuous spectrum of light astonished me; lightbulbs in my classroom and the sunlight reflecting off the playground were all enclosing a secret rainbow, a hidden world right in front of me.

Finally, in high school, I got the chance to study physics formally. I was amazed by how one could accurately predict the trajectory of objects with simple calculations. With this newfound knowledge, I attempted to construct a rocket and predict its motion, though I nearly set my garage on fire while testing my potassium chlorate fuel mixture. At this point, I was certain I wanted to study and research physics at a high level, and I was fortunate enough to be accepted into a research institute like Northwestern University and receive the scholarships I needed to attend. Once there, ***I was invited to apply to the Integrated Science Program, a selective program that aims to develop a highly interdisciplinary understanding of the sciences***, allowing students to study a mix of STEM courses from organic chemistry to quantum mechanics, while creating an accelerated path towards particular interests. However, it was not all sunshine and rainbows. Adjusting to both the rigor and the fast pace was a challenge at first, but I was stubborn enough not to give up and kept studying. As I continued to learn more through my classes and research, I always kept my grandfather's quote in the back of my mind, which remains one of my biggest motivations to pursue a career in physics and research at the highest level.

Intellectual Merit

My first exposure to precision measurement came from Professor Brian Odom, who taught my freshman fall seminar. I became deeply interested in the prospects of Atomic, Molecular, and Optical (AMO) physics after he told me about his past research at Northwestern's Center for Fundamental Physics (CFP) on trapping molecular ions as a potential precision search for time-varying fundamental constants. It was the first time I was exposed to the idea of tabletop experiments for fundamental physics. I was astonished by the capabilities of these new platforms and their ability to address the same questions as massive facilities like CERN or LIGO. Learning about Professor Odom's brand new project to grow doped cryogenic rare-gas crystals, I was ecstatic about the prospect of building a precision measurement platform from the ground up and joined his lab right away. Institutional knowledge in the Odom Lab, however, was lost due to a hiatus, so I started a temporary internship with Professor Tim Kovachy to build and transfer these skills into Professor Odom's lab.

I worked in Professor Kovachy's lab my freshman spring and summer, after first getting machine shop training that winter. They were developing tests to search for new short-range forces that can arise from beyond Standard Model physics using precision atom interferometry with strontium (Sr). Key to this endeavor is the ability to laser cool Sr in a magneto-optical trap. For this, I worked to develop a high-temperature strontium vapor cell for laser locking under the guidance of a PhD student. ***I designed and machined a stainless steel reservoir, including thermal isolation stages, to hold the Sr and implemented custom Nichrome mica heaters to reach temperatures up to*** $\mathbf{580^{\circ}C}$—well beyond the 500°C required to achieve the Sr optical density required. ***I also set up an ultra-high vacuum (UHV) chamber for outgassing tests*** on my heaters

and ***designed the optical layout for absorption spectroscopy***. During my time with Professor Kovachy, I developed foundational skills in SolidWorks design, thermal engineering, UHV chamber assembly, and spectroscopy, all of which were crucial to bring into Professor Odom's lab.

Starting in fall 2023, I began applying my newly acquired skills in Professor Odom's lab, ***helping design the primary UHV chamber to grow and study these doped cryogenic rare-gas crystals.*** These crystals are incredibly promising for fundamental physics and precision measurement, as they contain orders of magnitude more atoms or molecules than standard traps, enabling greatly improved statistical sensitivity. However, coherence times are currently limited, believed to be due to crystal inhomogeneity, so we are developing a system to study the growth of these crystals, creating a tunable, modular growth setup that allows X-ray and optical analysis of our crystals. Several considerations were required during chamber design, including compatibility with cryogenic and X-ray diffractometers, optical access, and the introduction of rare-gas and dopants. The last of which became my primary area of focus as ***I designed and implemented a highly collimated and tunable dual-species Rb–Ar beam source*** to transport our dopants and rare-gas $\frac{1}{2}$ m to our 1 cm growth substrate, utilizing a nozzle array of 231, 100 μm-diameter microcapillaries.

Initial characterization on Rb confirms a half-angle divergence of $\alpha_{1/e} = 0.48 \pm 0.1°$ corresponding to a capture efficiency of $68 \pm 7\%$ and Rb atom flux of 1.44×10^{10} atoms/s at the growth substrate. My current Ar gas handling system can control flow between $\sim 10^{15}$ and 10^{18} Ar atoms/s, and once implemented with my Rb source, should be able to grow doped crystals with dopant densities around 10^{16}–10^{18} dopants per cubic cm of rare-gas at rates of tens of micrometers per hour, consistent with previous experiments and confirming my initial design requirements. Additionally, a custom-designed set of Nichrome mica heaters and temperature sensors set the Rb temperature, allowing variable control of Rb flux, while a nozzle heater prevents Rb build-up and can maintain temperatures over 400°C for a week straight.

To characterize my beam, I had to both assemble a custom UHV characterization chamber and set up an Rb absorption spectroscopy system from the ground up, as seen in Fig. 1. I recommissioned an old ECDL and configured frequency modulation control to sweep over the Rb^{85} D_2 $F = 3 \rightarrow F' = 4$ resonant transition. I had to set up all required optical components, including mode shapers, collimating lenses, Faraday rotators to prevent back reflection, single-mode PM optical fibers, and an Rb reference cell setup to ensure resonant spectroscopy. By fitting absorption spectra of my beam from this setup and doing in-depth data analysis in Python, I was able to extract the values reported above. ***I am continuing to explore the accessible phase space of the beam and characterize achievable growth regimes, all of which will become part of my senior thesis and a planned instrumentation paper this coming spring.*** During my time in Professor Odom's lab, ***I have helped build a precision measurement platform from the ground up.*** The skills I learned in Professor Kovachy's lab were directly applied as I shared my experience in UHV and optics with an incoming PhD student and mentored other undergraduates as they joined the lab with the experiences and skills I had developed. I have had the privilege of ***representing the Odom Group with a poster presentation to my peers at the 2025 Midwestern Cold Atoms Workshop***, both sharing my dual-species beam work and the general progress our lab has made towards doped cryogenic rare-gas crystals. Additionally, my research has led me to become a ***2025 semi-finalist for***

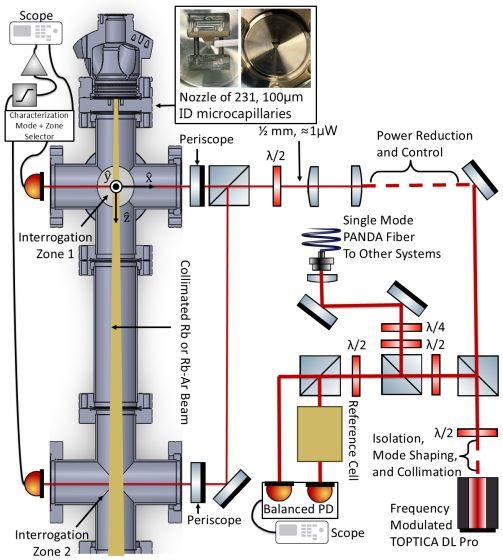


Figure 1: Simplified schematic of my dual-species beam characterization setup, built entirely by me from the ground up

the Fletcher Undergraduate Research Grant Prize award and to receive the ***2025 Junior Career Award in Physics and Astronomy,*** given to the one junior who demonstrates exceptional academic achievement and research potential at Northwestern University. Beyond lab work, my strong desire to continue developing an understanding of precision measurement and quantum science has led me to pursue a graduate course on Quantum Electronics and, currently, a graduate sequence on Advanced Topics in AMO Physics. Additionally, I both applied for and attended the ***2025 Johns Hopkins Summer School on Quantum Sensing and Precision Measurement***. All of these experiences have cemented my interest in and desire to pursue a PhD in AMO physics, focusing on precision measurement and fundamental physics.

Broader Impacts

The Integrated Science Program has been one of the most valuable programs and peer groups I have developed during my time at Northwestern. Beyond developing academics, the program really focuses on creating a collaborative community of interdisciplinary scientists, ranging from environmental scientists to biomedical engineers and physicists. Many of the friends I have developed in the community I have gone to for advice in my lab, such as asking an ISP electrical engineer for help with a filter and amplifier for data acquisition, and an ISP mechanical engineer taught me SolidWorks for design work. I have also often helped my fellow ISP members by assisting them in preparing for our physics and math exams. Being able to perform *real science*, at such an early age, and with such an amazing and collaborative group has been a treasuring experience. Developing rewarding friendships and having invaluable connections across disciplines is absolutely essential in today's science.

Within Professor Odom's lab, I have become a senior student and a crucial part of developing the lab culture and the way we do research moving forward. I was one of the first students with UHV and optics experience in our lab, so I played a significant role in sharing these skills with both undergraduates and a current PhD student. I have also played an important role in mentoring our newest undergraduate, showing him around the lab, explaining our current progress and goals, and helping him identify which projects would be appropriate for him. As the community has been such a great mentor to me, I'm proud to now return the favor and be a source of open advice to them. I have also developed standard operating procedures for my lab on the use of my laser spectroscopy setup and on loading rubidium into my beam source for future use. Along with my planned senior thesis manuscript, I am working to make my characterization code, custom heaters, and beam-source designs available for open-source use so others can build on them.

Career Goals

I plan to pursue a PhD in Atomic, Molecular, and Optical Physics, focusing on using light–matter interactions to search for physics beyond the Standard Model. I am eager to continue working with Professor Brian Odom on the promising platform of doped cryogenic rare-gas crystals. My existing experience has given me a unique opportunity to get in early and continue growing this platform to the level of competitive measurement it can achieve. I would also be very excited to work with other researchers and projects on this platform; such as Professor Jaideep T. Singh at Michigan State, part of the EDM3 collaboration, to use these crystals as a search for non-zero electric dipole moments (EDM). It would also be great to work on other precision measurement platforms, such as Professor Xing Fan's $^{229}ThF^{+}$ Schiff moment search for charge–parity violation. Regardless of where the science and opportunities take me, I will continue to search for work in collaborative and interdisciplinary environments, such as those experienced at Northwestern and the Center for Fundamental Physics. Upon finishing graduate school, I hope to continue in academia, becoming a post-doctoral researcher at a major research institution, and eventually become a professor in the field of precision measurement, developing the most sensitive tests for physics beyond the Standard Model.